

Nicholas Hu

Houston, TX 77005 | (414) 345-0001 | nkh5@rice.edu | nkh5.github.io/website/ | linkedin.com/in/nichhu | U.S. Citizen

EDUCATION

Rice University, Houston, TX

BS in Electrical and Computer Engineering, Minor in Data Science

Honors: National Merit Scholar (Rockwell Automation) | Chevron Engineering Scholarship (sole ECE recipient)

May 2027

GPA: 3.65/4.0

SKILLS

Languages: Python, Java, Go, Kotlin, TypeScript, C/C++, SQL, Bash, MATLAB

Tools & Libraries: AWS, Docker, Kubernetes, PostgreSQL, Git/GitLab, Jenkins, pandas, XGBoost, LightGBM, PyTorch, ONNX

EXPERIENCE

Rockwell Automation

Software Development Intern

Milwaukee, WI

June 2026 – August 2026

- Implemented device deletion through an existing **REST API** in **TypeScript**, mapping responses to application models and handling invalid IDs, missing devices, and service failures; passed **GitLab/Jenkins CI** checks.
- Redesigned **Kotlin** validation to link errors to conflicting device settings and corrected device-to-network lookups; added **unit tests** and updated **browser end-to-end tests** to verify both changes.
- Took ownership of a senior developer's in-progress performance optimization, rebased the branch and resolved merge conflicts, and delivered the work **4 days ahead of schedule**.

Software Development Intern

June 2025 – August 2025

- Automated **3 recurring manual regression cases** with **TypeScript** browser tests, published them through **Jenkins**, and verified successful execution across test runs to reduce recurring manual testing.
- Diagnosed failures across a **22-test nightly regression suite**, tracing regressions to recent merge requests and validating fixes; contributed to raising the team's nightly pass rate from **87.6% to 91.3%** after fixes were merged.

Control and Firmware Engineering Intern

June 2024 – August 2024

- Built a **MATLAB/Simulink** workflow that integrated visual model comparisons into **GitLab**, reducing manual comparison work and earning adoption by **2 additional engineering teams**.
- Evaluated motor-control software through **200 automated tests** and established a performance baseline with **32 efficiency tests** across motor loads and speeds.

Rice University ECE & Baylor College of Medicine

ML Student Researcher

Houston, TX

November 2024 – Present

- Built a **Python/XGBoost** pipeline to predict depression from wearable and smartphone data in a **26-participant, 90-day study**; achieved **0.74 AUC** at baseline using leave-one-subject-out cross-validation.
- Analyzed clinical, quality-of-life, and wearable data to identify a statistically significant link between limb-threat severity and foot circulation; coauthored a **Journal of Vascular Surgery** publication.

PROJECTS

GoldBench (Quantitative Forecasting Benchmark) | *Python, LightGBM, PyTorch, ONNX*

May 2026

- Built a reproducible **Python** benchmark comparing **5 model families** across **5 years of gold futures data** and **28 time-ordered evaluation periods**; found stronger predictive signal for volatility than market direction.
- Converted a **LightGBM** model to **ONNX**, reducing model size by **34%** and making predictions **2.8 times faster** while preserving identical outputs.

CourtOps (Pickleball Event Management) | *Go, PostgreSQL, Docker, AWS*

September 2026 – Present

- Built and used a **Go/PostgreSQL web application** to manage check-in, court assignments, and results for a **23-player pickleball event** organized with Rice Taiwanese Association and Vietnamese Student Association.
- Prevented duplicate court and player assignments using **PostgreSQL transactions, row locking, and unique indexes**; verified that **8 simultaneous assignment requests** produced exactly one match.
- Deployed on **AWS EC2** using **Docker Compose** and HTTPS, restricted organizer access through **Google OpenID Connect**, and verified database recovery from encrypted backups stored in **Amazon S3**.

LEADERSHIP

OwlSat (Rice CubeSat Club) | *Hardware Team Lead*

September 2023 – Present

- Lead a **7-member hardware/software team** integrating CubeSat subsystems; reviewed simulations with the communications lead to verify **1.6 Wh of energy margin per orbit**, preserving plans for multiple ground stations.

Rice IEEE | *Class Representative*

August 2023 – Present

- Led outreach and operations for the club's first professional-development speaker series, bringing Samsung, HPE, and Y Combinator speakers to campus and reaching **63 students (97% of the ECE graduating class)**.

RELEVANT COURSEWORK

Introduction to Database Systems, Machine Learning Concepts, Artificial Intelligence, Linear Algebra for Data Science, Data Science Tools and Models, Probability & Statistics